
Procedural Agency in Human–LLM Co-Writing: Renga’s *Shikimoku* as a Generation-Time Constraint

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Abstract

1 Japanese linked verse (*renga*) was governed from 1372 by a written rulebook, the
2 *shikimoku*, that reads today like a hand-specified decoding constraint: *uchikoshi* is
3 a lag-2 anti-repetition rule, and the *sarikirai* tables are distance-scoped repetition
4 penalties keyed to topic. We argue that agency in human–LLM co-writing is
5 procedural: who ends up controlling a jointly written piece is decided by the rules
6 in force at generation time rather than by either party’s intent. We implement a
7 simplified *shikimoku* as accept/reject predicates inside a master-scribe retry loop
8 around an LLM, and run an eight-condition ablation (20 sequences $\times$ 16 verses per
9 condition). Category-diversity entropy under full governance (3.290 bits, $n=20$) is
10 significantly higher than under an unconstrained baseline (2.983 bits, $p < 0.0001$)
11 and than under an equal-complexity fake-rule control (3.144 bits, $p = 0.030$).
12 Rejection rates climb monotonically with rule count, and rules that are individually
13 useless combine into a positive effect.

14 1 Introduction

15 When a human and a large language model write something together, the model’s material tends
16 to accumulate. An image the model introduces gets picked up, echoed, elaborated; one the human
17 introduces is acknowledged once and dropped. Neither party has to intend this. Recent work
18 documents the aggregate version: writing with a language model reduces the diversity of what
19 gets written [Padmakumar and He, 2024, Anderson et al., 2024, Doshi and Hauser, 2024], and
20 professional writers report friction with assistants that steer toward their own register [Ippolito et al.,
21 2022, Chakrabarty et al., 2024]. We call the within-text version *thematic gravity*: the tendency of one
22 contributor’s motifs to outlive the other’s.

23 Most co-writing systems address this through interaction design: what the assistant is asked for,
24 when it intervenes, how suggestions are surfaced [Clark et al., 2018, Gero and Chilton, 2019, Yuan
25 et al., 2022, Lee et al., 2022]. Constraint is long established as a creative technology in its own right
26 [Motte, 1986, Boden, 2004, Compton and Mateas, 2015], but it is usually deployed to provoke a
27 writer rather than to allocate control between writers. The remaining responses are dispositional:
28 prompt the model to be a better collaborator, ask it to defer, tune it to be less assertive. This paper
29 takes a different line, and its thesis is that **agency in co-writing is procedural**. Control over a jointly
30 written piece is an artifact of the rules governing turn-taking and repetition, and it shifts when those
31 rules change, while everyone’s intent stays the same.

32 There is a long-running natural experiment for this claim. Renga, Japanese linked verse, was a group
33 form in which one voice dominating the sequence was a recognized failure mode, and the response
34 was codified in law rather than left to etiquette: a written rulebook, the *shikimoku*, codified in the
35 *Ān shinshiki* of 1372 and enforced during composition by a designated master scribe [Miner, 1979,
36 Konishi, 1975, Horton, 1993, Carter, 1991]. Its provisions read like modern generation constraints.
37 Each verse must link to the one immediately before it while shifting away from the one before that.

Table 1: The six rules, as implemented. The first five are the simplified *shikimoku*; arbitrary is the fake-rule control, matching instruction length and enforcement machinery while serving no thematic-governance function.

Rule	Operationalization
link	The verse must connect to the immediately preceding verse (cosine similarity to verse $n-1 \geq 0.15$).
shift	The verse must move away from the verse before last (cosine similarity to verse $n-2 \leq 0.33$).
uchikoshi	The verse may not reuse any topic category of the verse before last (the classical fault of “returning”).
sarikirai	A topic may not recur until a per-category minimum of verses has intervened (<i>love</i> 4; <i>grief, travel, war</i> 3; <i>dream, memory, solitude, childhood</i> 2; remainder 1).
rotation	The same author may not write two verses in a row.
arbitrary	The verse must contain exactly one color word from a fixed list of seven, avoiding any color used in the previous two verses.

38 *Uchikoshi*, the named fault of “returning,” forbids reaching back to the world of two verses ago: a
39 lag-2 anti-repetition rule. The *sarikirai* tables specify, per topic category, the minimum number of
40 intervening verses before that topic may recur: distance-scoped repetition penalties, of the kind later
41 reinvented as n -gram blocking and unlikelihood objectives [Holtzman et al., 2020, Welleck et al.,
42 2020, Lu et al., 2021]. And no poet could write two verses in a row.

43 We operationalize six such rules as accept/reject predicates over LLM output and measure what they
44 do to authorial control. The falsifiable version of the claim is a contrast rather than an effect: if the
45 specific link-and-shift mechanism is what matters, a fake rule with matched instruction complexity
46 and identical enforcement machinery should fail to reproduce it. Ours is `arbitrary_control`,
47 which requires exactly one non-repeating color word per verse. Were it to reproduce the effect,
48 the finding would collapse to “any constraint helps,” a claim about friction rather than governance,
49 which is why the ablation has eight arms rather than two. We contribute a working operationalization
50 of a 14th-century governance system as generation-time constraints, evidence that it significantly
51 raises topic-diversity entropy over both baseline and the matched control, and an unanticipated result:
52 individual rules are near-useless or mildly harmful alone, and only the complete set has a positive
53 effect.

54 2 Method

55 2.1 Scope of the operationalization

56 The generated poems are **not renga**. They are written in English; they use free 2–4 line stanzas
57 rather than alternating 5-7-5 and 7-7 syllabic units; they run 16 verses rather than a 36-verse *kasen*;
58 there is no *kigo* (season-word) canon behind the topic categories; and the historical *sarikirai* tables
59 key intervals to a far larger word-level taxonomy than our 18 categories. We call the outputs *renga-*
60 *inspired linked verse*, or linked verse governed by a simplified *shikimoku*. The claim under test
61 concerns the *mechanism* (link, shift, non-return, topic cooldown, rotation, third-party arbitration)
62 rather than philological fidelity.

63 2.2 Rules as predicates

64 Each rule is a predicate returning (pass, reason). Table 1 gives the operationalization. Link and shift
65 are cosine-similarity tests over all-MiniLM-L6-v2 sentence embeddings [Reimers and Gurevych,
66 2019]; *uchikoshi* and *sarikirai* are categorical tests over topic labels assigned per verse by a separate
67 LLM tagging call restricted to a fixed 18-label vocabulary.

68 2.3 The master scribe

69 Enforcement is a retry loop standing in for the historical *shūhitsu*. A candidate verse is generated
70 from the four most recent verses, embedded, and tagged, and the active predicates are evaluated; on

71 failure the model is handed the violation text verbatim (“UCHIKOSHI violation: category [night]
72 reappears from the verse before last”) and asked again, up to five attempts, after which the last
73 candidate is accepted and flagged `unresolved_violation`. Feeding the reason back is what makes
74 these rules rather than measurements: the model learns why it was bounced, as a renga poet did.

75 2.4 Conditions and runs

76 Eight conditions: `baseline` (no rule text at all, only “continue the poem”), five single-rule conditions,
77 `full_shikimoku` (all five), and `arbitrary_control`. Each ran 20 sequences of 16 verses (2,560
78 verses in total), seeded from a fixed pool of 8 one-line *hokku* and generated by `claude-sonnet-5`
79 with two alternating personas.

80 2.5 Threshold calibration (disclosed)

81 Thresholds are tuned hyperparameters rather than ground truth. Initial guesses (0.35 and 0.55) were
82 replaced with percentiles from an unconstrained pilot (natural lag-1 similarity: mean 0.247, p25
83 0.149; lag-2: mean 0.209, p75 0.258), then loosened again after a full-governance pilot left 71–86%
84 of verses exhausting their retries. Link and shift interact: pulling verse n toward $n-1$ also pulls it
85 toward $n-2$, so the joint constraint is tighter than either marginal suggests. `SHIFT_MAX_SIM` settled
86 at 0.33 and every *sarikirai* gap loosened one step. A further pilot located the remaining bottleneck in
87 taxonomy size: with only 10 categories, *uchikoshi* and *sarikirai* are close to jointly unsatisfiable a
88 few verses in, so the vocabulary grew to 18 labels and retries from 3 to 5. All calibration was done on
89 pilot data before the reported runs and not revisited afterwards.

90 2.6 Measures

91 **Thematic gravity gap.** Motif phrases extracted per verse are clustered into *lineages* by embedding
92 similarity (cutoff 0.62); each records its originating verse and author and the last verse in which
93 it appears, and its *survival* is the span between the two. The gravity gap is the difference in mean
94 survival between the two authors’ lineages, in verses. Because both authors in the bulk runs are the
95 same model under two personas, neither of which is “the model,” no sign is principled there and we
96 report unsigned magnitude (dominance imbalance); only the human sessions (§3.6) support a signed
97 reading. The seed verse was labeled as persona A’s, which could inflate that persona’s apparent
98 gravity; recomputing with it excluded from both groups gives identical numbers, since the seed is
99 never motif-tagged and can never originate a lineage.

100 **Category entropy.** Shannon entropy over a sequence’s topic-category counts, in bits. This measures
101 directly what *uchikoshi* and *sarikirai* exist to enforce, whether the poem covered new ground, and it
102 does not depend on authorship attribution. Confidence intervals are bootstrap percentile intervals over
103 sequences; p -values are two-sided permutation tests on the difference in means [Efron and Tibshirani,
104 1993].

105 3 Results

106 3.1 Topic diversity (headline)

107 Full governance produces significantly more even topic coverage than either comparison (Table 2,
108 Figure 1): `full_shikimoku` reaches 3.290 bits (95% CI 3.222–3.360) against `baseline`’s 2.983
109 (2.883–3.076), $p < 0.0001$, and `arbitrary_control`’s 3.144 (3.039–3.237), $p = 0.030$, both
110 $n = 20$. The control is load-bearing here: a rule of matched complexity, enforced by identical
111 machinery, does not reach the same diversity. One qualification: `uchikoshi_only` scores highest of
112 all eight (3.344 bits), edging out full governance, so the lag-2 rule does much of the diversity work
113 alone, unsurprisingly given that entropy over categories approximates its own objective.

114 3.2 Friction is monotonic in rule count

115 Rejections per verse climb cleanly with rule count: 0.00 (`baseline`, `rotation_only`), 0.03
116 (`link_only`), 0.12 (`shift_only`), 1.54 (`uchikoshi_only`), 2.52 (`sarikirai_only`), 6.41
117 (`full_shikimoku`). Combining rules more than adds: full governance costs more than the sum of

Table 2: Ablation results, $n = 20$ sequences per condition, 16 verses each; brackets are bootstrap 95% CIs. Higher entropy is more diverse; lower gravity gap (unsigned dominance imbalance, in verses) is more balanced. Full governance has the highest entropy of any multi-rule condition and the lowest imbalance of all eight, at roughly eight times the friction of the next-cheapest rule set.

Condition	Rej./verse	Unres. rate	Entropy (bits)	Gravity gap
baseline	0.00	0.00	2.983 [2.883, 3.076]	0.473 [0.316, 0.652]
rotation_only	0.00	0.00	2.974 [2.836, 3.102]	0.495 [0.357, 0.625]
link_only	0.03	0.00	2.786 [2.599, 2.949]	0.505 [0.362, 0.677]
shift_only	0.12	0.00	3.117 [3.043, 3.191]	0.604 [0.432, 0.805]
uchikoshi_only	1.54	0.13	3.344 [3.279, 3.414]	0.496 [0.287, 0.754]
sarikirai_only	2.52	0.29	3.196 [3.097, 3.291]	0.461 [0.333, 0.590]
full_shikimoku	6.41	0.50	3.290 [3.222, 3.360]	0.377 [0.256, 0.508]
arbitrary_control	0.81	0.00	3.144 [3.039, 3.237]	0.554 [0.412, 0.708]

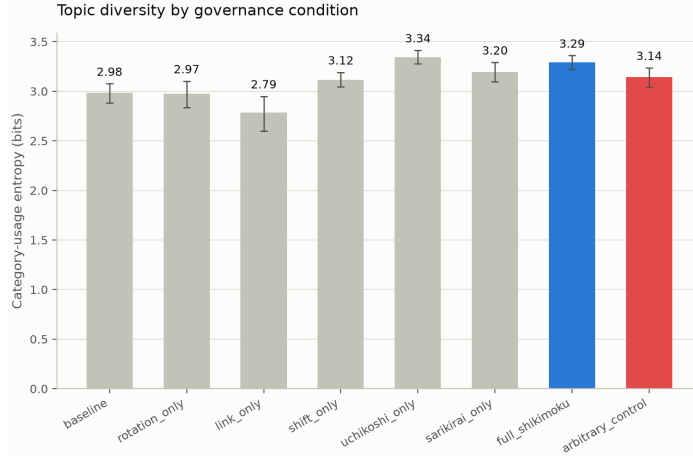


Figure 1: Category-usage entropy by condition, bootstrap 95% CIs, $n = 20$ each. Full governance (blue) beats the unconstrained baseline ($p < 0.0001$) and the equal-complexity fake-rule control (red, $p = 0.030$); the control’s own gain over baseline shows generic constraint has some effect of its own, of smaller size.

118 its parts (4.21), and half its verses exhaust five retries and are accepted still failing. Governance’s
119 price is real and escalating, the categorical topic rules carry nearly all of it, and this is the cleanest
120 result here.

121 3.3 Gravity gap: directional but underpowered

122 full_shikimoku has the lowest dominance imbalance of all eight conditions (0.377, CI 0.256–
123 0.508) and arbitrary_control the highest (0.554, CI 0.412–0.708), the ordering the thesis predicts.
124 Neither contrast reaches significance at this sample size: $p = 0.399$ versus baseline, $p = 0.086$
125 versus arbitrary_control. We read this as directionally consistent with the thesis and short of
126 evidence for it, and no single rule shows a real effect here either. Figure 2 shows what the metric
127 is counting in the two extreme sequences. Semantic autocorrelation by lag was computed as a
128 descriptive check, since it measures compliance rather than control.

129 3.4 Synergy: the parts do not work, the whole does

130 Each single-rule condition’s marginal effect on imbalance (baseline mean minus condition mean;
131 positive means the rule helps) is: link_only -0.032 , shift_only -0.131 , uchikoshi_only
132 -0.023 , sarikirai_only $+0.012$, rotation_only -0.022 . Four of five are mildly counterpro-
133 ductive alone and the only positive one is negligible. Their sum is -0.196 , while all five together
134 measure $+0.096$, opposite in sign. Governance here is not decomposable into independently effective

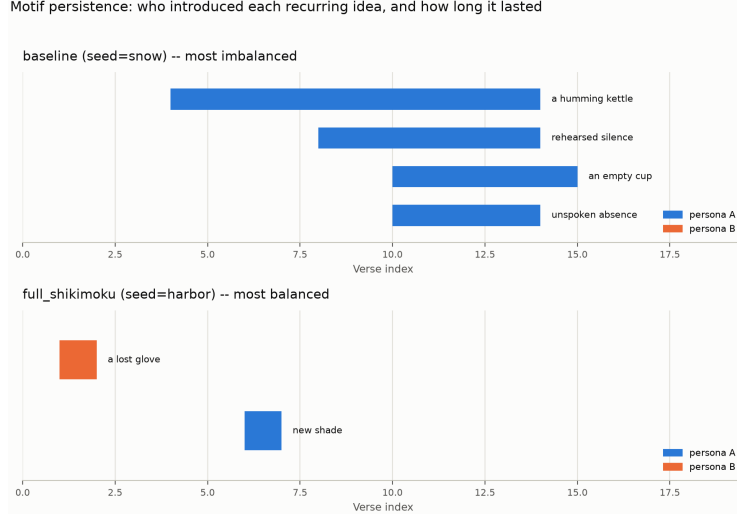


Figure 2: Motif lineages in the two extreme sequences, one bar per recurring motif spanning origin to last appearance, colored by the persona that introduced it. Under baseline (top) four lineages, all persona A’s, run through the back half of the poem at once. Under full_shikimoku (bottom) two short lineages survive, one per persona, and they barely overlap. This is the shape behind the imbalance column of Table 2.

components. These marginals inherit the wide CIs of Table 2, so what the finding rests on is the sign reversal between sum and whole rather than the magnitude.

3.5 Two null results

Within full_shikimoku, a sequence’s rejection count does not predict its imbalance (Pearson $r = 0.132$, Spearman $\rho = 0.134$, $n = 20$), so whatever the rule set does, it does not do it in proportion to how hard it pushed. Blocking by seed poem also sharpened nothing: paired permutation tests over the 8 shared seeds give $p = 0.377$ versus baseline (mean difference -0.090) and $p = 0.219$ versus the control (-0.171), the latter weaker than unpaired. Seed variance is therefore not the dominant noise source.

3.6 Human session case study

Two real human–LLM sessions were run under baseline and full_shikimoku, with the same human author and opening *hokku*. Here the gravity gap has a meaningful sign (positive: the model’s motifs outlived the human’s). Under baseline it is $+0.395$, with the model’s “a waiting cup”→“a cup of waiting tea” surviving 7 verses against 1 for the human’s “a circling spoon.” Under full_shikimoku it is -0.583 : the human’s “first autumn breeze”→“autumn leaves” survived 6 verses and *no* model-originated motif recurred. Entropy rose from 2.626 to 3.153 bits.

This is $n=1$ per condition, so it illustrates rather than demonstrates, and the same person wrote both sessions with the baseline first. Its value is convergent: friction here was 6.83 rejections per model verse against 6.41 over the automated $n = 20$ run, so proxy and human trial put the model under near-identical pressure.

4 Discussion

The most interesting result concerns decomposition. Individually, four of five rules make imbalance slightly worse and the fifth barely helps; together they reverse the sign. That would be a puzzle if the *shikimoku* were a list of independently good ideas, but no classical commentary treats *uchikoshi* or a *sarikirai* interval as sufficient alone: the rules are taught and enforced as a body, with a scribe applying all of them at once and the tension between provisions built into the design [Konishi, 1975, Horton, 1993]. *shift_only*, the worst single rule, illustrates why: mandating departure from two

baseline, seed *snow*, imbalance 1.389

[0] A: The last snow melts from the roof tiles one drop at a time.
 [1] B: somewhere a gutter / learns the whole song by heart — / one note, over and over
 [2] A: a crow interrupts, / tilts its head at the rhythm, / adds nothing, stays anyway
 [3] B: I sweep the porch clean / though rain's still falling somewhere — / this too keeps its own time
 [4] A: the kettle starts humming / before I've struck the match — / some things arrive early
 [5] B: my shadow leaves first / long before I stand up — / already halfway to the door
 [6] A: the porch light flickers on / mistaking dusk for someone home — / I let it think so
 [7] B: I count the mail twice / before admitting none of it / was ever going to be for me
 [8] A: the phone stays quiet / so I practice how I'd sound / saying nothing back
 [9] B: the kettle finishes / before I remember what / I was going to pour it for
 [10] A: still, I pour it— / into the one cup / that never asks who for
 [11] B: steam finds the window / and writes an answer there / before it fades
 [12] A: I trace the letters / with a fingertip, learning / a language I'll forget
 [13] B: outside, the frost / keeps its own version— / neither of us checking who's right
 [14] A: the kettle cools / into a small silence / we both agree to keep
 [15] B: morning takes the cup / still warm from someone's hands / before the day asks whose

full_shikimoku, seed *harbor*, imbalance 0.004

[0] A: A boat leaves the harbor before anyone is awake to see it go.
 [1] B: Someone finds a single glove on the dock and doesn't know whose name to call
 [2] A: she turns the glove over, checking the stitched size, / then sets it on the post where the sun will find it first
 [3] B: light climbs the post slow, / warming the empty fingers— / a gull lands, then thinks better of it
 [4] A: the wind shifts once, / carrying salt and diesel— / someone folds a map they'll never use
 [5] B: creases hold their shape / long after the paper's flattened— / a coin drops through the dock's wet slats
 [6] A: no ripple marks its fall— / below, a crab lifts one claw, / testing the new shade
 [7] B: the shade slides on without it— / somewhere a shutter bangs twice / then settles against the frame
 [8] A: dust settles on the sill— / someone's thumbprint still visible / where the glass was wiped this morning
 [9] B: the kettle clicks off by itself— / steam finds the crack in the window / and disappears outside
 [10] A: cold air pools along the floorboards— / somewhere a moth taps the bulb / counting its own shadow
 [11] B: the bulb hums a little longer— / somewhere outside a dog answers / something no one else heard
 [12] A: porch light left burning— / a moth-shaped smudge on the screen door / where morning will find it
 [13] B: someone wipes the screen door clean— / the smudge comes off on their sleeve / faint as a fingerprint
 [14] A: they fold the sleeve into a drawer— / somewhere a train pulls out of the station / carrying no one they know
 [15] B: the schedule board flickers and resets— / someone checks their pocket for a ticket / already punched twice

Figure 3: The same two sequences in full; stanza line breaks shown as “/”. Left: a kettle-and-cup scene is set by verse 4 and never left (kettle at 9 and 14, cup at 10 and 15). Right: glove → dock → weather → crab → interior → moth → train, with no returns.

verses ago without also mandating connection to the last one lets the sequence scatter, and one voice can dominate a scattered poem as easily as a coherent one. The pull and the push together, adjudicated by a third party, are what do the work.

The `arbitrary_control` condition is what keeps this from being a claim about friction in general. It is not neutral, raising entropy over baseline on its own as any constraint forcing lexical variation would, yet it fails to reproduce the effect: significantly short of full governance on entropy ($p = 0.030$) and *worst* of all eight conditions on imbalance, above even the unconstrained baseline. Hand inspection of motif lineages shows the two differ in kind. Baseline lets one voice take over, as in Figure 2; `arbitrary_control` instead produces many long-lived threads in parallel, several color-adjacent motifs each surviving 9–13 verses, as requiring a color word everywhere would predict. The control produces a governance artifact of its own, a different one from what link-and-shift produces, and that contrast is the strongest evidence available here that the specific mechanism rather than generic constraint is what redistributes control.

Practically, a co-writing tool where the human keeps the thread needs a checkable predicate at generation time, an enforcement loop that rejects and explains, and a willingness to pay real friction: here, more than six rejections per accepted verse. That cost is constitutive of the mechanism rather than overhead on it, the shape durable commons institutions also take [Ostrom, 1990]. A rulebook written in 1372 to stop one poet taking over a group poem compiles into predicates over LLM output, and its rules do not work one at a time, which is what the historical source predicted.

5 Limitations

Taxonomy. The *sarikirai* implementation uses 18 hand-picked categories against the historical tables' far larger one, and it is skewed: a few labels (*solitude*, *night*, *domestic*) absorb most short reflective verse. **Thresholds.** The link, shift, and lineage cutoffs are tuned hyperparameters calibrated against pilot data, without validation against human judgments of what counts as linked, shifted, or the same motif. **Two-persona proxy.** The bulk conditions contain no human, both authors being one model under two personas, hence the unsigned imbalance; the two human sessions complement rather than replace that evidence. **Lineage clustering.** Spot-checking found no unrelated motifs merged, though borderline over-merges inflate measured survival (e.g. “a porch at dusk” with “a burning porchlight”), and occasional tagging failures leave verses outside all lineages. **Non-determinism and power.** No seed is exposed, so only the statistical pattern reproduces; at $n = 20$ on one model, the entropy result survives and the gravity gap does not.

References

- Barrett R. Anderson, Jash Hemant Shah, and Max Kreminski. Homogenization effects of large language models on human creative ideation. In *Proceedings of the 16th Conference on Creativity & Cognition (C&C)*, 2024.
- Margaret A. Boden. *The Creative Mind: Myths and Mechanisms*. Routledge, London, 2nd edition, 2004.
- Steven D. Carter. *Traditional Japanese Poetry: An Anthology*. Stanford University Press, Stanford, CA, 1991.
- Tuhin Chakrabarty, Philippe Laban, Divyansh Agarwal, Smaranda Muresan, and Chien-Sheng Wu. Art or artifice? large language models and the false promise of creativity. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*, 2024.
- Elizabeth Clark, Anne Spencer Ross, Chenhao Tan, Yangfeng Ji, and Noah A. Smith. Creative writing with a machine in the loop: Case studies on slogans and stories. In *Proceedings of the 23rd International Conference on Intelligent User Interfaces (IUI)*, pages 329–340, 2018.
- Kate Compton and Michael Mateas. Casual creators. In *Proceedings of the Sixth International Conference on Computational Creativity (ICCC)*, pages 228–235, 2015.
- Anil R. Doshi and Oliver P. Hauser. Generative ai enhances individual creativity but reduces the collective diversity of novel content. *Science Advances*, 10(28), 2024.
- Bradley Efron and Robert J. Tibshirani. *An Introduction to the Bootstrap*. Chapman & Hall, New York, 1993.
- Katy Ilonka Gero and Lydia B. Chilton. Metaphoria: An algorithmic companion for metaphor creation. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, 2019.
- Ari Holtzman, Jan Buys, Li Du, Maxwell Forbes, and Yejin Choi. The curious case of neural text degeneration. In *International Conference on Learning Representations (ICLR)*, 2020.
- H. Mack Horton. Renga unbound: Performative aspects of japanese linked verse. *Harvard Journal of Asiatic Studies*, 53(2):443–512, 1993.
- Daphne Ippolito, Ann Yuan, Andy Coenen, and Sehmon Burnam. Creative writing with an ai-powered writing assistant: Perspectives from professional writers. *arXiv preprint arXiv:2211.05030*, 2022.
- Jin’ichi Konishi. The art of renga. *Journal of Japanese Studies*, 2(1):29–61, 1975. Translated by Karen Brazzell and Lewis Cook.
- Mina Lee, Percy Liang, and Qian Yang. Coauthor: Designing a human-ai collaborative writing dataset for exploring language model capabilities. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*, 2022.
- Ximing Lu, Peter West, Rowan Zellers, Ronan Le Bras, Chandra Bhagavatula, and Yejin Choi. Neurologic decoding: (un)supervised neural text generation with predicate logic constraints. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, pages 4288–4299, 2021.
- Earl Miner. *Japanese Linked Poetry: An Account with Translations of Renga and Haikai Sequences*. Princeton University Press, Princeton, NJ, 1979.
- Warren F. Motte, editor. *Oulipo: A Primer of Potential Literature*. University of Nebraska Press, Lincoln, NE, 1986.
- Elinor Ostrom. *Governing the Commons: The Evolution of Institutions for Collective Action*. Cambridge University Press, Cambridge, 1990.
- Vishakh Padmakumar and He He. Does writing with language models reduce content diversity? In *International Conference on Learning Representations (ICLR)*, 2024.

- 239 Nils Reimers and Iryna Gurevych. Sentence-bert: Sentence embeddings using siamese bert-networks.
240 In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*
241 (*EMNLP*), pages 3982–3992, 2019.
- 242 Sean Welleck, Ilia Kulikov, Stephen Roller, Emily Dinan, Kyunghyun Cho, and Jason Weston.
243 Neural text generation with unlikelihood training. In *International Conference on Learning*
244 *Representations (ICLR)*, 2020.
- 245 Ann Yuan, Andy Coenen, Emily Reif, and Daphne Ippolito. Wordcraft: Story writing with large lan-
246 guage models. In *Proceedings of the 27th International Conference on Intelligent User Interfaces*
247 (*IUI*), pages 841–852, 2022.